Q $y = \operatorname{arccsc} x$ $\quad \frac{dy}{dx}?$

A $\operatorname{arccsc}(x) = y$

$\csc(y) = x$

$-\csc(y)\cot(y) \cdot \frac{dy}{dx} = 1$

$$\frac{dy}{dx} = \frac{1}{-\csc(y)\cot(y)}$$

We can replace $\csc(y)$ with x but what about $\cot(y)$?

$\sin^2 y + \cos^2 y = 1$

$\frac{\sin^2 y}{\sin^2 y} \quad \frac{\cos^2 y}{\sin^2 y} = \frac{1}{\sin^2 y}$

$1 + \cot^2 y = \csc^2 y$

$\cot^2 y = \csc^2 y - 1$

$\cot(y) = \pm\sqrt{\csc^2 y - 1}$

$\cot(y) = \sqrt{\csc^2 y - 1}$

$$\frac{dy}{dx} = \frac{1}{-\csc(y)\sqrt{\csc^2 y - 1}}$$

$$\frac{dy}{dx} = \frac{1}{-x\sqrt{x^2 - 1}}$$